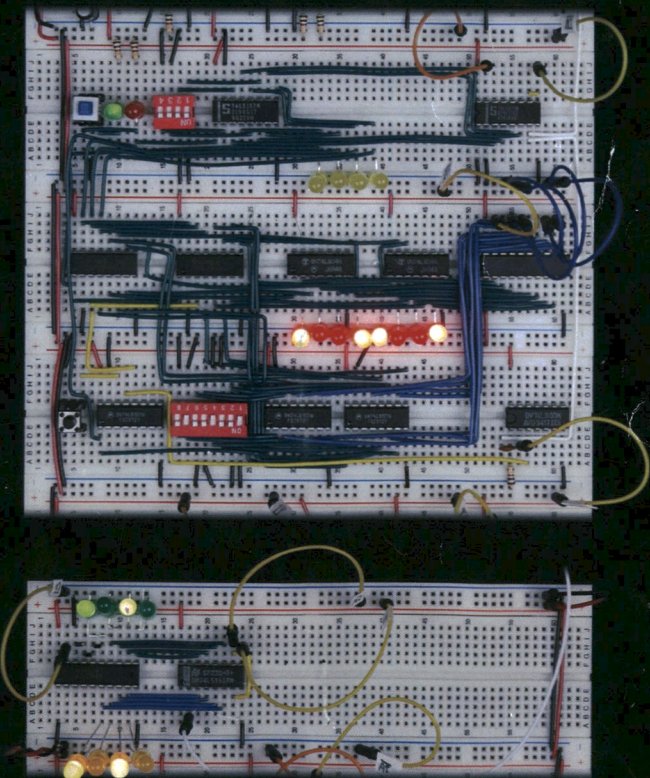
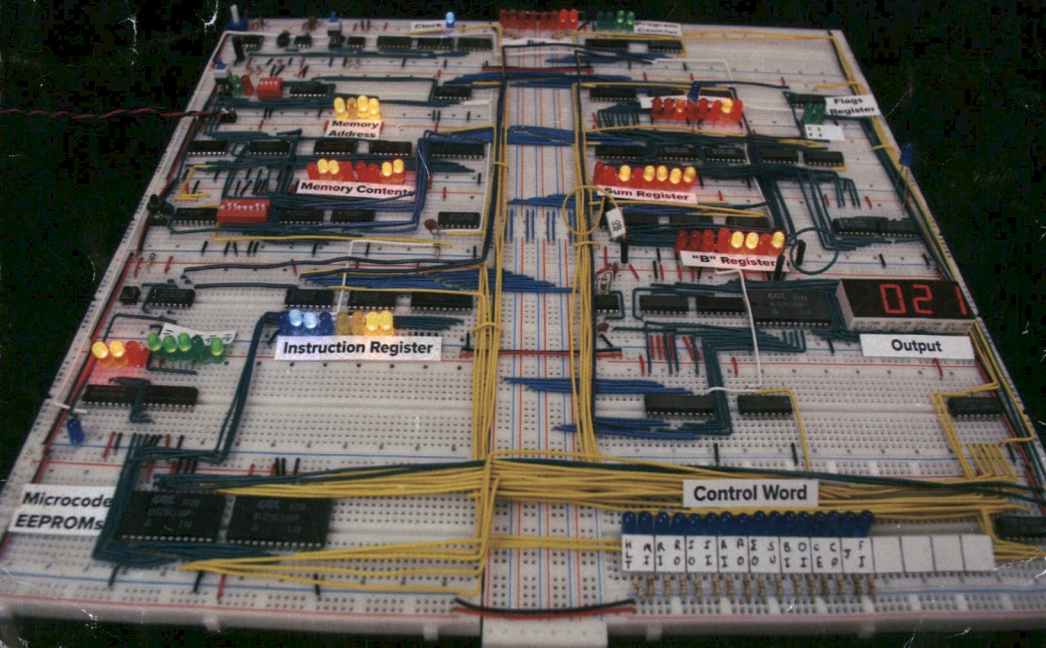


8-bit breadboard computer

Kit 3 of 4

RAM and Program Counter

www.eater.net
youtube.com/beneater



Ben Eater

Welcome!

This is the third kit in a set of four that, along with my series of videos on YouTube, will help you develop a deep understanding of how programmable CPUs really work. I hope when you're done, you'll know enough to extend the computer design and make it your own.

With this third kit, you'll build the RAM module and program counter.

The random access memory (RAM) stores the program the computer is executing as well as any data the program needs. Our breadboard computer uses 4-bit addresses which means it will only have 16 bytes of RAM, limiting the size and complexity of programs it can run. This is by far its biggest limitation.

The program counter (PC) counts in binary to keep track of which instruction the computer is currently executing.

Where are the instructions to build this kit?

Surprise! This booklet doesn't actually contain the full instructions for building the kit. This kit is designed as a companion to my series of YouTube videos on building a computer from scratch. You should watch the videos and follow along yourself. I hope that by *not* having a detailed "paint-by-numbers" instruction book, you'll learn more about how the circuits really work. But rest assured, the videos are quite detailed and should contain everything you need to be successful:

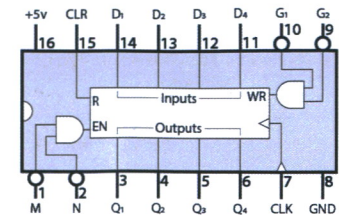
The video instructions for this kit are at

www.eater.net/8bit/ram

www.eater.net/8bit/pc

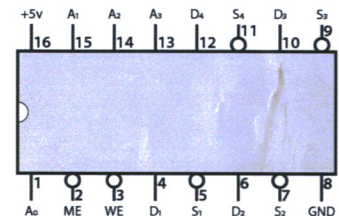
74LS173: 4-bit D register

This register stores 4 bits. You write data to it by holding G1 and G2 low and pulsing the clock.



74189: 64-bit RAM

This static RAM holds 16 4-bit words. After applying a binary number to the four address inputs, data can be read or written to the memory.



74LS245: 8-bit bus transceiver

This allows 8-bit data from A to B or B to A depending on the DIR input. The EN input disables the device so that A and B are isolated.

